



Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

Certified Lean Six Sigma White Belt (CLSSWB)

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Conducted by

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1	1 July 2026	Initial release — CLSSWB Learner Guide for the one-day Lean Six Sigma awareness course.	Dr. Alfred Ang
2	19 July 2026	Guide rebuilt from the single-source content module and restructured to follow the DMAIC roadmap end to end, with exactly one hands-on lab per DMAIC phase (5 labs). All labs unified under one continuous scenario (the BrewBean Cafe morning rush) so each lab output feeds the next. Content simplified to White Belt awareness depth: sigma-level and DPMO calculation, MSA, FMEA, value stream mapping, Kano analysis, weighted solution-selection matrices and SPC control limits removed and replaced with recognition-level explanations; quick-reference section changed from formulas to the DMAIC roadmap; glossary aligned to the tools actually taught.	Dr. Alfred Ang

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Introduction

This Learner Guide accompanies the course Certified Lean Six Sigma White Belt (CLSSWB) (C142), conducted by Tertiary Infotech Academy Pte Ltd. This is a one-day Lean Six Sigma awareness course. It follows the DMAIC roadmap end to end — Define, Measure, Analyze, Improve, Control — with one hands-on lab in each of the five phases, and provides step-by-step instructions for every lab.

The course content is grounded in the body of knowledge published by The Council for Six Sigma Certification (CSSC) in 'Six Sigma: A Complete Step-by-Step Guide', so what you learn here matches the recognised White Belt standard.

As a White Belt you are being prepared to CONTRIBUTE to an improvement team — to understand the language, recognise waste, and support the tools your team uses. You are not expected to lead a project or perform the statistical analysis; those belong to the Yellow, Green and Black Belt levels.

Every lab uses one continuous scenario — the BrewBean Cafe morning rush, where customers queue up to 15 minutes and orders are sometimes made wrong. By the end of the day your five lab outputs form a complete improvement story: customer requirements and problem statement, process map and waste tally, root cause analysis, a selected countermeasure with a pilot plan, and a control plan.

Course Learning Outcomes

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

Skills Framework Alignment

Abilities

Knowledge

Before You Start

What you need

- A laptop with a spreadsheet application (Microsoft Excel, Google Sheets or LibreOffice Calc), or simply pen and paper.
- A browser, for the interactive problem-solving tools used in the Analyze lab.
- The course slides and this Learner Guide, downloaded from <https://lms-tms.tertiaryinfotech.com>.
- A work process of your own to think about — the tools apply far better when the example is real.

The interactive problem-solving toolkit

Two browser-based tools are used during the labs. No installation or licence is required.

- 5 Whys — build and share a 5 Whys chain: <https://alfredang.github.io/5whys/>
- Fishbone Diagram — build an Ishikawa cause-and-effect diagram: <https://alfredang.github.io/fishbone/>

How the five labs fit together

- There are five labs — exactly one for each phase of DMAIC: Define, Measure, Analyze, Improve and Control.
- All five labs build on the same BrewBean Cafe scenario, so each lab's output becomes the next lab's input.
- Every lab is completed by everyone, and all five build one continuous improvement project.

Conventions used in every lab

- Each lab states its objective, the deliverable you produce, the steps, and a check to confirm you are done.
- Tables shown in the steps can be built in a spreadsheet or on the worksheet provided.
- Where a lab uses an online tool, the tool URL is shown with the step.
- Keep every lab output — they combine into your final improvement package and are your revision material.

FOUNDATIONS — Lean Six Sigma Foundations (20%)

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

Key concepts

- What is Quality — Quality is meeting the customer's requirements — not just being free of defects.

- What is Lean — Lean removes waste so work flows faster to the customer.
- What is Six Sigma — Six Sigma reduces variation so results become consistent and predictable.
- Lean Six Sigma — The two combined: faster flow AND fewer defects, decided by data not opinion.
- Belt roles — White, Yellow, Green, Black and Master Black Belt — who does what on a team.
- The DMAIC roadmap — Define, Measure, Analyze, Improve, Control — the five-step improvement path.

DEFINE — Define — Understand the Problem (20%)

Voice of the Customer · CTQ · Problem statement · SMART goal · Project charter

Key concepts

- Voice of the Customer — Ask customers what they need, and record it in their own words.
- Critical to Quality — Turn each customer need into something specific you can measure.
- Problem statement — What is wrong, where, since when, and how big — never the solution.
- SMART goal — Specific, Measurable, Achievable, Relevant and Time-bound.
- Project charter — A one-page summary of the problem, goal, scope and team.
- Scope — Agreeing what is in and out keeps a small project small.

Lab 1 — Define — Customer Requirements and the Problem Statement [Core]

Objective: Identify customer requirements and write a clear problem statement (A1, A2).

Goal: Start the BrewBean Cafe improvement the way every real project starts: find out what the customer actually wants, turn that into something measurable, and state the problem clearly enough that everyone agrees on it. The most common reason improvement projects fail is that the team never agreed what the problem was.

What you'll build

A VOC-to-CTQ table, a written problem statement, a SMART goal and a simple scope table. (Tools and techniques: Voice of the Customer, CTQ, problem statement, SMART goal, project scope.)

Step-by-step

1. Read the BrewBean Cafe scenario and underline every customer complaint you can find.
2. Write down five Voice of the Customer statements in the customer's own words (e.g. 'I'm late for work because the queue is too slow').

3. Turn each VOC statement into a measurable CTQ requirement — add a number and a unit (e.g. 'served within 5 minutes').
4. Write the problem statement using the template: what is wrong, where, since when, and how big. Do NOT write a solution.
5. Check your problem statement against the SMART test and rewrite the goal so it is Specific, Measurable, Achievable, Relevant and Time-bound.
6. Fill the in-scope / out-of-scope table so the team knows where the project stops.

Check your work

Every CTQ has a number and a unit, your problem statement contains no solution, and your goal has a metric, a baseline, a target and a date.

Note: The full worksheet for this lab is in labs/lab-01-*.md.

MEASURE — Measure — See What Is Really Happening (20%)

Process mapping · SIPOC · Types of data · Data collection · The eight wastes

Key concepts

- Process mapping — Draw the steps as they really happen, not as the manual says.
- SIPOC — A one-page overview: Suppliers, Inputs, Process, Outputs, Customers.
- Types of data — Discrete data you count; continuous data you measure on a scale.
- Data collection — Agree what to record, who records it, and when — before you start.
- Check sheets — A simple tally form is the easiest reliable way to collect data.
- The eight wastes — DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

Lab 2 — Measure — Map the Process and Spot the Waste [Core]

Objective: Describe how the process runs and identify the eight wastes in it (A4, K2).

Goal: You cannot improve what you cannot see. Walk the BrewBean Cafe morning rush from the customer joining the queue to the drink being handed over, draw it as it really happens, and tally where time and effort are being lost. This is the phase that turns opinions about the problem into facts about the problem.

What you'll build

A SIPOC overview, a step-by-step process map with timings, and a completed waste tally sheet. (Tools and techniques: SIPOC, process mapping, types of data, check sheets, the eight wastes (DOWNTIME).)

Step-by-step

7. Agree the start and stop points of the process — start: customer joins the queue; stop: customer receives the drink.
8. Complete the SIPOC template: Suppliers, Inputs, Process (5-7 steps only), Outputs, Customers.
9. List the process steps in order and write the observed time next to each one.
10. Mark each step as value-adding (the customer would pay for it) or not value-adding.
11. Classify the data you collected: which figures are discrete (counted) and which are continuous (measured)?
12. Walk the process again with the DOWNTIME check sheet and tally every waste you observe against its type.
13. Identify which single waste type you tallied most often, and which process step it happens at.

Check your work

Your SIPOC has all five columns filled, every process step has a time, and every waste on your tally sheet is tagged to one of the eight DOWNTIME types.

Note: The full worksheet for this lab is in labs/lab-02-*.md.

ANALYZE — Analyze — Find the Cause (20%)

Root cause · 5 Whys · Fishbone diagram · Pareto chart · Variation

Key concepts

- Symptom vs cause — Treating the symptom makes the problem come back; treating the cause does not.
- Root cause — The cause that, once removed, stops the problem recurring.
- 5 Whys — Keep asking 'why?' until you reach something you can actually act on.
- Fishbone diagram — Sort possible causes into Manpower, Method, Machine, Material and Measurement.
- Pareto chart — Roughly 80% of the problem usually comes from 20% of the causes.
- Variation — Some variation is normal in the process; some has a specific assignable cause.

Lab 3 — Analyze — Find the Root Cause [Core]

Objective: Identify likely causes using 5 Whys and Fishbone analysis (A3, K2).

Goal: You now know WHERE the delay happens. This lab asks WHY it happens. Most teams jump straight from a symptom to a solution and end up fixing the wrong thing — the queue is long, so hire more staff — when the real cause was something else entirely. Here you drill past the symptom to a cause you can actually act on.

What you'll build

A completed 5 Whys chain, a Fishbone diagram with causes sorted by category, and a shortlist of likely root causes. (Tools and techniques: Symptom vs cause, 5 Whys, Fishbone (5M) diagram, Pareto chart, common vs special cause variation.)

Step-by-step

14. Write the symptom from your Lab 2 findings at the top of the 5 Whys template (e.g. 'customers wait 15 minutes').
15. Ask 'why does that happen?' five times, each answer becoming the next question. Stop when you reach something you can act on.
16. Draw the Fishbone diagram and write the problem in the head of the fish.
17. Brainstorm possible causes onto the five bones: Manpower, Method, Machine, Material and Measurement.
18. Read the supplied Pareto chart of BrewBean complaint types and state which few causes account for most of the problem.
19. Decide for each shortlisted cause whether it is common cause (happens every day) or special cause (a one-off event).
20. Shortlist the two or three causes best supported by the data you collected in Lab 2.

Check your work

Your 5 Whys chain ends in something the team can actually act on, every Fishbone cause sits under one of the five categories, and each shortlisted cause is backed by an observation, not an opinion.

Note: The full worksheet for this lab is in labs/lab-03-*.md.

IMPROVE — Improve — Fix the Cause (10%)

Generating solutions · 5S · Mistake proofing · Standard work · Piloting

Key concepts

- Generating solutions — Brainstorm against the proven cause — never against the symptom.
- Choosing a solution — Compare ideas on impact, effort and risk before committing.
- 5S — Sort, Set in order, Shine, Standardise, Sustain — a tidy workplace for physical or digital work.
- Mistake proofing — Poka-Yoke makes the error hard or impossible to make.
- Standard work — Write the better method down so everyone does it the same way.
- Piloting — Try the change small first, so mistakes stay cheap.

Lab 4 — Improve — Choose and Pilot a Countermeasure [Core]

Objective: Describe improvement actions that address the identified cause (A5, K2).

Goal: A cause is only worth finding if something changes because of it. Take the root causes you shortlisted in Lab 3, generate countermeasures against them, and pick one that is small enough to try next week. The discipline here is to solve the CAUSE you proved, not the symptom you started with.

What you'll build

A brainstormed countermeasure list, an impact/effort grid, one selected countermeasure and a simple pilot plan. (Tools and techniques: Brainstorming, impact/effort screening, 5S, mistake proofing (Poka-Yoke), standard work, piloting.)

Step-by-step

21. Write your top root cause from Lab 3 at the top of the sheet — every idea must address THIS cause.
22. Brainstorm at least six countermeasures without judging any of them yet.
23. Check whether 5S would help: is anything at the counter hard to find, out of place or untidy?
24. Check whether mistake proofing would help: could the wrong order be made impossible rather than merely discouraged?
25. Plot every idea on the impact/effort grid and pick one from the high-impact, low-effort corner.
26. Write the improved method as standard work — short numbered steps anyone on shift could follow.
27. Write a one-week pilot plan: what changes, who runs it, what you will measure, and how you will know it worked.

Check your work

Your selected countermeasure clearly addresses the root cause from Lab 3 (not the symptom), and your pilot plan names a measure, an owner and a review date.

Note: The full worksheet for this lab is in labs/lab-04-*.md.

CONTROL — Control — Hold the Gain (10%)

Control plan · Visual management · SOPs · Team huddles · Handover

Key concepts

- Why Control matters — Without Control, processes quietly drift back to the old way.

- Control plan — What we measure, the target, how often, who owns it, and what to do if it slips.
- Visual management — Make performance visible so problems are noticed the same day.
- Standard operating procedures — Written instructions that lock in the improved method.
- Team huddles — Short regular stand-ups catch problems while they are still small.
- Handover — Give the improved process back to the people who run it every day.

Lab 5 — Control — Hold the Gain and Hand Over [Core]

Objective: Describe the control actions that sustain an improvement (A5, K2).

Goal: Most improvements work for a fortnight and then quietly disappear. Control is what stops that happening. Close the BrewBean Cafe story by deciding what will be measured from now on, who owns it, what happens when it slips, and how the improved process is handed back to the people who run it every day.

What you'll build

A completed control plan, a visual management board sketch, and a one-page summary of the whole DMAIC story. (Tools and techniques: Control plan, visual management, SOPs, team huddles, handover, one-page summary.)

Step-by-step

28. Choose the ONE measure that tells you the improvement is still working (e.g. average wait time at 8:00am).
29. Complete the control plan template: measure, target, how often it is checked, and who owns it.
30. Write the reaction plan — the exact steps to take when the measure misses target.
31. Sketch a simple visual board that would make this measure visible to the whole shift.
32. Turn your Lab 4 standard work into a short SOP the cafe could actually pin up.
33. Plan a five-minute daily huddle: who attends, what is reviewed and how long it lasts.
34. Summarise the whole project on one page — problem, cause, countermeasure, result and control — and name who you hand it over to.

Check your work

Your control plan names a measure, a target, a frequency and an owner; your reaction plan says what to DO when it slips; and your one-page summary tells the complete DMAIC story from problem to handover.

Note: The full worksheet for this lab is in labs/lab-05-*.md.

Quick Reference — The DMAIC Roadmap

What each phase asks, and what it delivers

- Define — 'What problem are we solving, and for whom?' Delivers the Voice of the Customer, the CTQ requirements, a problem statement, a SMART goal and an agreed scope.
- Measure — 'How big is the problem, really?' Delivers a SIPOC, a process map, a data collection plan and a baseline of how the process performs today.
- Analyze — 'Why is this happening?' Delivers the shortlisted root causes, supported by 5 Whys, a Fishbone diagram and the data collected in Measure.
- Improve — 'What change will actually fix it?' Delivers a selected countermeasure, standard work for the new method, and a pilot plan.
- Control — 'How do we make the gain stick?' Delivers a control plan, visual management, an SOP and a handover to the process owner.

A note on Six Sigma measurement

You will hear practitioners refer to a process's 'sigma level' and to DPMO (Defects Per Million Opportunities) — the measure Six Sigma uses to describe how often a process produces a defect. The well-known Six Sigma target is 3.4 defects per million opportunities, which is why the method carries that name.

As a White Belt you are expected to RECOGNISE these terms and understand what they describe. Calculating DPMO and converting it to a sigma level is done at Yellow Belt and above, and is not assessed on this course.

Quick Reference — The Eight Wastes (DOWNTIME)

- D — Defects: output that fails the requirement and must be corrected or redone.
- O — Overproduction: producing more, or earlier, than the customer needs.
- W — Waiting: work or people idle, waiting for the next step, an approval or information.
- N — Non-utilised talent: skills and ideas of people not being used.
- T — Transport: unnecessary movement of materials, work items or information between places.
- I — Inventory: work in progress, backlogs and queues sitting between steps.
- M — Motion: unnecessary movement of people, or switching between systems and screens.
- E — Extra-processing: doing more work to the output than the customer requires or values.

Consolidating What You Learned

- Re-read your own lab outputs; together they form one complete DMAIC improvement story.
- Be ready to define quality, Lean, Six Sigma and Lean Six Sigma, and explain how they differ.
- Be ready to describe the belt roles and say where a White Belt contributes to an improvement team.
- Be ready to name the eight wastes (DOWNTIME) and give a workplace example of each.
- Be ready to explain each DMAIC phase, what it delivers and which tools belong to it.
- Be ready to translate a VOC statement into a measurable CTQ requirement.
- Be ready to write a problem statement that describes the problem without naming a solution.
- Be ready to explain how the Fishbone diagram and 5 Whys are used together to find a root cause.
- Be ready to describe what a control plan must contain to hold an improvement in place.
- Re-work the labs from memory — being able to produce the tools unaided is the real test of understanding.

Glossary

- **Lean** — A method to maximise customer value by systematically identifying and removing waste.
- **Six Sigma** — A data-driven method to reduce variation and defects; the target is 3.4 defects per million opportunities.
- **Lean Six Sigma** — The combined method — Lean improves speed and flow, Six Sigma improves consistency and accuracy.
- **DMAIC** — Define, Measure, Analyze, Improve, Control — the Six Sigma improvement roadmap.
- **PDCA** — Plan, Do, Check, Act — a lighter improvement cycle used for small, fast improvements.
- **VOC** — Voice of the Customer — customer needs and expectations expressed in the customer's own words.
- **CTQ** — Critical to Quality — a specific, measurable requirement translated from a VOC statement.
- **SIPOC** — Suppliers, Inputs, Process, Outputs, Customers — the macro 'as-is' process map.
- **Muda** — The Japanese term for waste — any activity that consumes resource but creates no customer value.
- **DOWNTIME** — Mnemonic for the eight wastes: Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

- **Value-added (VA)** — An activity the customer would be willing to pay for because it changes the product or service.
- **Non-value-added (NVA)** — An activity that consumes resource but adds nothing the customer values — pure waste.
- **Defect** — An output that fails to meet the CTQ requirement.
- **Common cause variation** — Natural, always-present variation built into the process; addressed by changing the process.
- **Special cause variation** — Unusual variation traceable to a specific assignable event; addressed by investigating that event.
- **Pareto principle** — The 80/20 rule — roughly 80% of effects come from 20% of causes.
- **5 Whys** — Repeatedly asking 'why' to drill from a symptom down to an actionable root cause.
- **Fishbone (Ishikawa) diagram** — A cause-and-effect diagram organising candidate causes into categories such as the 5Ms.
- **5M categories** — Manpower, Method, Machine, Material, Measurement — standard Fishbone categories.
- **DPMO** — Defects Per Million Opportunities — the defect-rate measure Six Sigma uses. Recognise the term; calculating it is a Yellow Belt skill.
- **Yield** — The percentage of output produced without defects.
- **Sigma level** — A yardstick for process performance derived from DPMO; the Six Sigma target is 3.4 DPMO.
- **Discrete data** — Data you obtain by counting, such as the number of orders made wrong.
- **Continuous data** — Data you obtain by measuring on a scale, such as waiting time in minutes.
- **Check sheet** — A simple structured tally form used to collect process data consistently.
- **5S** — Sort, Set in order, Shine, Standardise, Sustain — a Lean method for organising the workplace.
- **Poka-Yoke** — Mistake proofing — designing the process so an error is difficult or impossible to make.
- **Standard work** — The documented current best-known method for performing a task.
- **Pilot** — A small-scale trial of a change, run before full rollout so mistakes stay cheap.
- **Control plan** — The document naming the metric, target, monitoring method, frequency, owner and reaction plan.
- **Reaction plan** — The agreed steps to take when a controlled measure misses its target.
- **Visual management** — Making process performance visible at a glance, so problems are noticed the same day.

- **SOP** — Standard Operating Procedure — written instructions that lock in the improved method.